

Key interests

Themes Computational social science, Complex systems, Climate crisis, Science communication

Methods Mathematical modelling, Agent-based simulation, Data science

Academic carrer

Postdoc
2024 – at [Complexity Science Hub](#) in Vienna, Austria
project: [Collective Adaptation](#)
main collaborators: [Mirta Galesic](#), [Henrik Olsson](#)

Ph.D.
2024 in **(Computational) Sociology** at [ZMT Bremen](#) and [Constructor University](#)
on ‘[The complex effects of distorted social perceptions on opinions about climate change](#)’
supervisor: [Agostino Merico](#);
main collaborators: [Paul E. Smaldino](#), [Marijn A. Keijzer](#)

Master (M.Sc.)
2020 in **Computer Simulations in Science and Engineering (COSSE)** at [TU Berlin](#) and [KTH Stockholm](#)
thesis in socio-economic modelling: ‘[A spatially explicit agent-based model of human-resource interaction on Easter Island](#)’ at [ZMT Bremen](#) (supervisor: [Agostino Merico](#))

Master (M.Sc.)
2018 in **(Climate) Physics** at [University of Potsdam](#) and [PIK](#)
thesis in climate modelling: ‘[Estimating global warming from anthropogenic heat emissions – conceptual and numerical modelling approaches](#)’ (supervisors: [Georg Feulner](#) & [Maria A. Martin](#))

Bachelor (B.Sc.)
2016 in **Physics** at [FAU Erlangen-Nuremberg](#)
thesis on simulating the detection of astro-particles (supervisor: [Uli Katz](#))

Scientific contributions

Selected publica-
tions **Steiglechner P**, Smaldino PE, & Merico A (2025). How opinion variation among in-groups can skew perceptions of ideological polarization [Preprint]. *SocArXiv*. <https://doi.org/10.31235/osf.io/yau9s>

Steiglechner P, Keijzer MA, Smaldino PE, Moser D, & Merico A (2024). Noise and opinion dynamics: How ambiguity promotes pro-majority consensus in the presence of confirmation bias. *Royal Society Open Science*, 11 (4), 231071. <https://doi.org/10.1098/rsos.231071>

Steiglechner P, Smaldino PE, Moser D, & Merico A (2023). Social identity bias and communication network clustering interact to shape patterns of opinion dynamics. *Journal of The Royal Society Interface*, 20 (209), 20230372. <https://doi.org/10.1098/rsif.2023.0372>

Steiglechner P & Merico A (2022). Spatio-temporal patterns of deforestation, settlement, and land use on easter island prior to European arrivals. In: Rull V & Stevenson C (eds) The prehistory of Rapa Nui (Easter Island): Towards an integrative interdisciplinary framework. *Developments in Paleoenvironmental Research*, vol 22. Springer, Cham. https://doi.org/10.1007/978-3-030-91127-0_16

Ongoing work **Steiglechner P**, Poulsen VM, Olsson H, Galesic M (in preparation). Adaptive belief networks.

Steiglechner P, Poulsen VM, Olsson H, Galesic M (in preparation). How we perceive opinions and evaluate political differences.

Batzke M, **Steiglechner P**, Lorenz J, Edmonds B, & Kalvas F (in preparation). From cognitive coherence to political polarization: An agent-based model of belief change.

Keijzer MA, Gelastopoulos A, Jochim A, Steinbrinker L, **Steiglechner P**, & Analytis PP (in preparation). From preference to dominance: The reinforcing and regressing effects of social influence.

Peer Review [Communication Research](#), [JASSS](#), [Springer](#)

Conference contributions Computational social science: [SSC 2024](#), [SSC 2023](#), [ODCD 2023](#), [SSC 2022](#)

Complexity science: [CCS 2022](#)

Climate science: [EGU General Assembly 2019](#), [DPG 2019](#)

Teaching (at university level) _____

Co-supervisor for master thesis on modelling individual decision-making under social influence (Uni Bremen, 2023) and on modelling the coral-algae symbiosis under stress (Uni Bremen, 2021)

Lecturer ‘Data Science Tools in python’ at Constructor University in collaboration with [Jan Lorenz](#) (2023)

Guest lectures ‘Agent-based model of the Covid-19 pandemic’ as part of the course ‘Modelling and analysis of complex systems’ by Agostino Merico at Constructor University (2020, 2022, 2023)

Teaching assistant ‘Modelling and analysis of complex systems’ by Agostino Merico at Constructor University (2020, 2022)

[winter schools](#) on ‘Mathematical modelling of ecological and socioeconomic systems’ at [AIMS Senegal](#) (2020, 2022) and at [ACECoR in Ghana](#) Ghana (2025)

 ‘Theoretical physics 2: Electrodynamics’ by Florian Marquardt at FAU (2015 – 2016)

Further research experience and education _____

Research stays at UC Merced and the Santa Fe Institute (2023, three months)

 at Imperial College London and Grantham Institute (2016, three months)

 policy briefing paper ‘[Novel policy tools to assess the environmental impacts of air pollutants](#)’ with [Apostolos Voulgarakis](#) and [Alyssa Gilbert](#)

Science communication [interview](#) on noise and ambiguity in communication (2024)

[radio interview](#) on global warming through anthropogenic heat emissions (2024)

 ‘[Wissenschaft persönlich](#)’—interview (2023)

 ‘[Science goes PUBLIC](#)’—informal science talks in pubs, by the Bremen city administration (2023)

 ‘[Book-a-scientist](#)’—one-on-one conversations with the general public, by the Leibniz Association (2020–2022)

Other relevant skills and interests _____

Programming Python (advanced), R, NetLogo (all beginner)

Politics and student organisations active in ‘Students for Future’ (2019), the Green university group (2012–2017), and the physics student association (2012–2017)

 member of the student councils on faculty and university level (2013/14)

Teaching taught intensive math courses preparing high school students for A-level exams ([abiturma](#), 2018–2024)